FROZEN — May 2026 EMG HDC Baseline (M16 + M17)

This is the commit-ready baseline snapshot. Config, seeds, encoding, and protocol are frozen so the June RTL can target identical numbers.

Frozen statement

> **May EMG baseline = 90.30 % ± 0.13 (spatial) under protocol P-may2026**, using the binary BSC model at **D = 1024** (the model the 1024-bit RTL implements), on the Rahimi ICRC-2016 EMG data, averaged over 5 subjects and 5 seeds.

Headline numbers

Model	Config	Spatial	Spatiotemporal
Project baseline (BSC)	D=1024, XOR / permute / majority / Hamming	**90.30 % ± 0.13**	**89.25 % ± 0.89**
Parity anchor (MAP)	D=10000, multiply / add / cosine	90.36 %	96.04 %
Paper (Rahimi 2016)	D=10000, MAP	90.8 %	97.8 %

The project's binary 1024-bit model matches the bipolar D=10000 anchor on the spatial task (90.30 % vs 90.36 %) and the published 90.8 %. The ±0.13 std over 5 seeds shows the spatial baseline is essentially seed-independent.

Per-seed (project baseline, D=1024)

Seed	Spatial	Spatiotemporal
1	90.12 %	89.77 %
2	90.24 %	89.79 %
3	90.29 %	90.16 %
4	90.51 %	87.72 %
5	90.35 %	88.81 %
mean ± std	**90.30 % ± 0.13**	**89.25 % ± 0.89**

Frozen configuration (`config/emg_baseline.json`)

- **Dataset:** `HDC-EMG/dataset.mat` (5 subjects, 4 channels, 5 classes, enveloped EMG, 21 levels, `int(value)` quantization). GPLv3.
- **Project model:** binary `{0,1}`; bind = **XOR**; permute = **cyclic shift**; bundle = **thresholded majority (tie → 0)**; similarity = **Hamming/popcount**; **D = 1024**. RTL: `xor_permute_top.sv`, `permute_stage.sv`, `bundle_unit.sv`, `popcount_am.sv`.
- **Parity anchor:** bipolar `{-1,+1}`; multiply / add / cyclic-shift / cosine; conditional-add training (cos < 0.9); D = 10000.
- **Protocol P-may2026:** train fraction 0.25 (first 25 % per class, shuffled with rng = seed+100); test = full per-subject sequence; best n-gram per subject S1=4 S2=4 S3=3 S4=5 S5=4; spatiotemporal downsample 250 (S5=50); metric = per-subject accuracy, mean over 5 subjects.
- **Seeds:** baseline seed = 1; std computed over seeds {1,2,3,4,5}.

One command to reproduce

```
cd python_ref
python run_emg_baseline.py                  # full: 5 seeds + MAP parity (~50 s)
python run_emg_baseline.py --quick --no-parity   # fast sanity (1 seed, ~7 s)
```

Outputs `results/emg_baseline.json` (the machine-readable snapshot) and prints the frozen one-liner above.

Status

- **M16 DONE** — config frozen in `config/emg_baseline.json` (seed, D=1024, encoding, protocol all fixed).
- **M17 DONE** — baseline frozen: 90.30 % ± 0.13 spatial under P-may2026; snapshot at `results/emg_baseline.json`; single rerun command above.

This closes the May milestone (lock the software golden path). Next (June): target these numbers from RTL — wire the Stage B encoder into ModelSim co-simulation and start the new modules (`bundle_unit.sv`, `popcount_am.sv`, `encoder_top.sv`, `hdc_stream_wrapper.sv`).